
COMPRESSIT

File Compression, Backup and Search Utility Engine

PROJECT	Project-Based Learning – Phase-I
TEAM ID	DSCPP-III-2026-T362
DOMAIN	Data Structures & OOP
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ROLE	Text Search, Inverted Index & Documentation

Individual Contribution Report

1. Project Overview

CompressIt is a C++ based file-management utility that brings compression, backup, duplicate detection and text search into one system. The project is intended to make file management more convenient while applying Data Structures and Object-Oriented Programming concepts in a practical application.

The project objectives include developing a single utility for file management, implementing the major file operations, applying DSA and OOP concepts, and evaluating compression, storage and search performance.

2. My Assigned Contribution

Area	Contribution
Text Search	Working on the text-search component of the utility.
Inverted Index	Organizing searchable word-to-file information for the search process.
Documentation	Documenting the search module and its connection with the complete system.

3. Text Search Module

The purpose of the text-search part is to help the user locate required text from the files being managed by CompressIt. The approach is to process the available text and keep useful search information in an organized form.

When a user provides a search term, the stored search information can be used to identify the relevant files. This becomes particularly useful when the number of files increases and repeatedly checking every file becomes inconvenient.

4. Inverted Index

An inverted index maintains a relationship between a word and the files in which that word occurs. Instead of keeping the search process dependent only on repeatedly scanning all files, the index provides an organized reference for the search term.

Search term	Files containing the term
backup	File1, File3
search	File2, File3

For example, if the word “backup” occurs in File1 and File3, the index can keep the relationship: backup → File1, File3. A search for “backup” can then refer to this stored relationship to obtain the matching files.

5. Working Flow of My Module

01	Select files	Choose the files that need to be searched.
02	Read & process	Read the relevant text and prepare it for indexing.
03	Build index	Store word-to-file relationships in the inverted index.
04	Search	Match the user's search term with the stored index.
05	Result	Return the matching file information.

6. DSA & C++ Concepts Connected to My Work

Hashing / Hash Table — Useful for organizing word-based search information.

Inverted Index — Maintains the relationship between searchable words and files.

Vector / STL — Can be used for storing and handling collections of search results.

File Handling — Required to read and process text files.

OOP Concepts — The complete project uses classes, objects, encapsulation, abstraction and related C++ concepts.

7. Documentation Contribution

The documentation work covers the search module, its working approach and its role in the overall CompressIt workflow. Clear documentation is useful for explaining the module during evaluation and for keeping the project understandable for the team.

The documentation also helps connect the implementation with the DSA and OOP concepts used in the project, making it easier to explain how the academic concepts are applied in the actual utility.

8. Integration with CompressIt

The text-search module is one part of the complete system. Compression is responsible for reducing storage usage, backup supports file preservation, duplicate detection helps identify repeated files, and the search module provides a way to locate required text.

The modules are planned to work together through the project's common file and metadata flow, resulting in a single utility rather than separate tools for each operation.

9. Contribution Summary

Component	My Focus
Text Search	Organized searching of required text in files.
Inverted Index	Word-to-file mapping used to support the search process.
Documentation	Clear explanation of the module and its integration.

10. Conclusion

My contribution to CompressIt is focused on the search side of the project. The Text Search module and Inverted Index are intended to make locating required text more structured, while the documentation explains the module and its role in the complete system. This work directly connects the project's practical requirements with the Data Structures and C++ concepts covered in the PBL course.

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